

温州理工學院
WENZHOU UNIVERSITY OF TECHNOLOGY

本科毕业设计（论文）

外 文 翻 译

题　目	基于 Vue 和 SpringBoot 的景点门票系统的设计与开发
专　业	计算机科学与技术（专升本）
班　级	23 计算机科学与技术（专升本）一
学生姓名	应韩桢
学　号	23219141158
指导教师	周旭立，李名标
职　称	助教、教授

温州理工学院教务处制

温州理工学院本科毕业设计（论文）外文翻译

一、 外文资料来源及题目（注：含作者、书名、杂志名或外文数据库名等，英文文章或段落标题，原文附后。）

作者：Junhui Song,Min Zhang,Hua Xue

标题：Design and Implementation of a Vue.js-Based College Teaching System

杂志名： International Journal of Emerging Technologies in Learning

来源：中国知网

二、 译成中文后题目(译文附后)：

基于 Vue.js 的高校教学系统的设计与实现

基于 Vue.js 的高校教学系统的设计与实现

摘要：互联网信息技术的普及从根本上改变了学习方式。利用互联网信息技术提高高校教学效果是当前的研究热点。本文从高校教学的实际需要出发，设计了一个基于各种互联网信息技术的高校教学系统。具体来说，前端是在轻量级的渐进式Vue.js框架下开发的，该框架依赖于模型-视图-视图模型；基于浏览器-服务器架构建立了系统的总体结构；系统功能通过HTML5、Node.js和数据库技术实现；在Boot strap下实现了移动终端和桌面之间的兼容性。系统测试表明，基于Vue.js的高校教学系统运行稳定，达到了设计目标，满足了用户需求。研究结果为大学教学的现代化和质量提供了重要的新见解

关键词-Vue.js 框架、HTML5、教学系统、设计与实现

1 如何使用此模板

1.1 将样式应用于现有图纸

目前，互联网和信息技术已经成为人们生活中不可或缺的技术工具。将互联网和信息技术融入教育，极大地促进了高校教学信息化改革。面对信息环境下传统课堂教学无法满足大学生需求的尴尬局面，高校建立了基于局域网或互联网辅助传统课堂教学，为师生学习交流提供平台，从而提高教学效果。国外信息技术和互联网技术的发展早于中国。因此，国外对教学体系的研究和开发相对成熟。美国拥有最广泛的网络教育，例如，在麻省理工学院（MIT），到 2006 年底，共享在线课程的数量已达到 2000 门。Blackboard 是美国公司开发的教学平台；它也是世界上使用最广泛的，为教师提供了强大的课程开发和教学环境，以及为学生提供良好的虚拟学习环境。Sakai 是美国多所大学联合开发的开源协作学习平台，旨在整合高校现有的组件和工具模块，构建在线教学管理。Moodle 是澳大利亚教师基于建构主义理论开发的一个教学辅助系统，由资源、论坛、课程管理等多个模块组成，并得到了许多机构的认可。尽管国内大学教学体系起步较晚，但发展迅速。4A 网络教学平台是中国最早开发的。该平台由教学资源管理、课程协同开发工具、教学管理系统和教学支持系统组成，增加了师生互动，极大地调动了学生的学习兴趣。清华大学建立的清华在线是各大学建立的网络平台中最成熟的一个。北京师范大学开发的 VClass 解决了传统教学中的许多繁琐问题，还可以为远程学习提供服务，实现教育资源的平衡。随着互联网信息技术的普及和智能手机，大学生已经习惯了获取知识和通过互联网学习相关技能。中国的大学教学体系发展迅速，但仍存在发展不平衡等问题层次、更新不及时、课程结构不完整、功能单一等。高校教学发展现状与研究现状分析本文试图研究使用 Vue.js 框架的大学教学系统的设计与实现 Vue.js 框架、HTML5、Node.js 等涉及的相关技术系统开发过程。然后，从实际教学需求出发分析了系统的各种需求，详细讨论了关键模块，并演示了具体的系统实现。最后，系统测试结果表明，基于 Vue.js 的学院教学系统运行稳定，达到了设计目标，满足了用户需求

2 系统相关技术

2.1 Vue.js框架

Vue.js 是 Web 应用程序中基于 MVVM 模式的最佳轻量级前端框架。图 1 显示了 MVVM 模式架构的示意图，其中 ViewModle 作为中间件负责功能和数据之间的通信

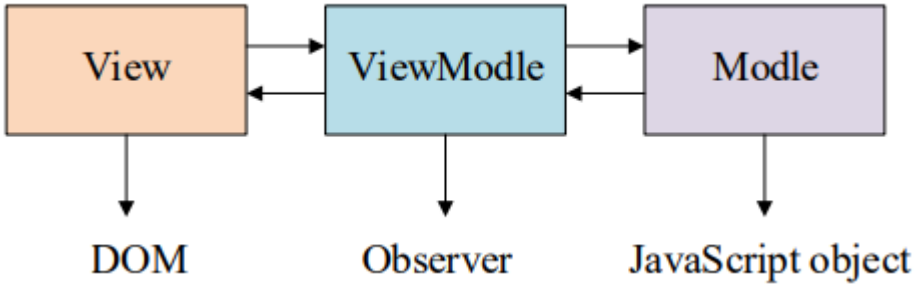


图 1.MVVM 模式架构图

Vue.js 核心库只关注功能层，因此不是一个通用的框架。Vue.js 可以通过简单的 API 实现响应式双向数据绑定，并快速构建用户界面。在 Vue.js 中，任何类型的应用程序接口都可以抽象为组件树。图 2 显示了组件系统的示意图。如果合理抽象组件，可以重用几个小组件来构建一个大系统，从而减少重复开发

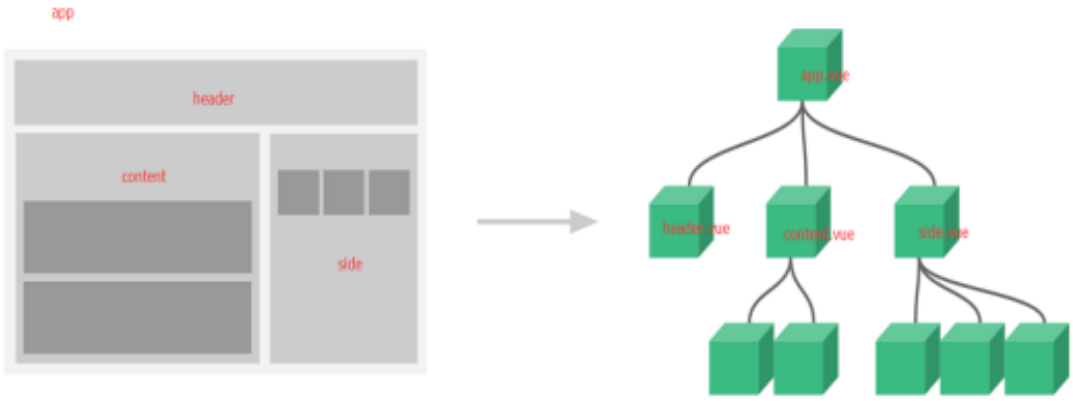


图 2.Vue.js 组件系统原理图

2.2HTML5

超文本标记语言（HTML）由“head”和“body”组成。用标记符号标记网页的每个部分是标准的标记语言[15]。HTML 被广泛使用，因为它允许浏览器通过点击 URL 指针（超链接）轻松访问新的网页。HTML5[16]是最新一代的 HTML 标准。基于 HTML，它具有更多新的有用功能，如性能和集成、web 多媒体、本地存储、语义功能等。这是近十年来 web 开发标准的最大飞跃

2.3Node.js

Node.js 是一个开发平台，允许 JavaScript 在服务器端运行，基于 JavaScript 的高性能、开源运行时环境[17]。它的本质是一个基于事件的框架，通过事件驱动机制防止 Web 请求的二次触发。本文使用 Node.js 事件驱动、异步阻塞 I/O、单线程特性进行系统后端设计[18]

3.基于 Vue.js 框架的高校教学系统的设计与实现

3.1 系统需求分析

系统的功能需求分析和非功能需求分析是软件开发的重要组成部分。功能需求分析是确定系统的主要功能和模块划分。非功能需求分析基于用户的需求和经验，以确保系统的安全性、稳定性、易维护性和可扩展性。本文主要研究系统的功能需求分析。

大学教学体系的设计是为了弥补传统课堂教学的局限性，即时间和空间的限制，以及师生之间缺乏互动交流，使学生能够根据自己的时间表和实践能力进行预习和课后复习。它还为教师和学生的交流搭建了一个互动平台。学生和教师是该系统的主要服务对象。此外，管理员负责通过综合管理确保系统的正常运行。因此，本文从学生的学习需求、教师的教学需求和管理员的管理需求三个方面分析了系统的功能需求。

学生：登录后，学生可以修改自己的信息，浏览课程列表，选择课程，观看教师提供的教学资源、课件和其他教学资源，在线学习或下载，查看教师发布的课程公告、作业公告和相关消息。他们还可以对选定的课程进行在线测试，查看成绩和论文分析，并通过论坛与教师和其他学生即时沟通。

教师：登录后，教师可以修改自己的用户信息，创建班级和课程，管理班级信息，发布相关教学资源和公告，安排学生作业，维护试题，查看学生的作业和在线测试分析，了解学生的学习情况，通过论坛与学生交流并回答他们的问题。

管理员：登录后，管理员可以分配用户角色，修改学生和教师的信息，管理系统中每个功能模块的信息

3.2 系统功能模块划分

根据系统需求分析，系统分为四个模块：登录功能、学习功能、论坛功能和测试功能，如图 3 所示。

登录模块：学生、教师和系统管理员可以通过系统登录功能进行注册和登录。登录后，他们可以修改个人信息，点击系统主页上的相应功能按钮进入每个功能模块，查看评论内容和自己的书签。

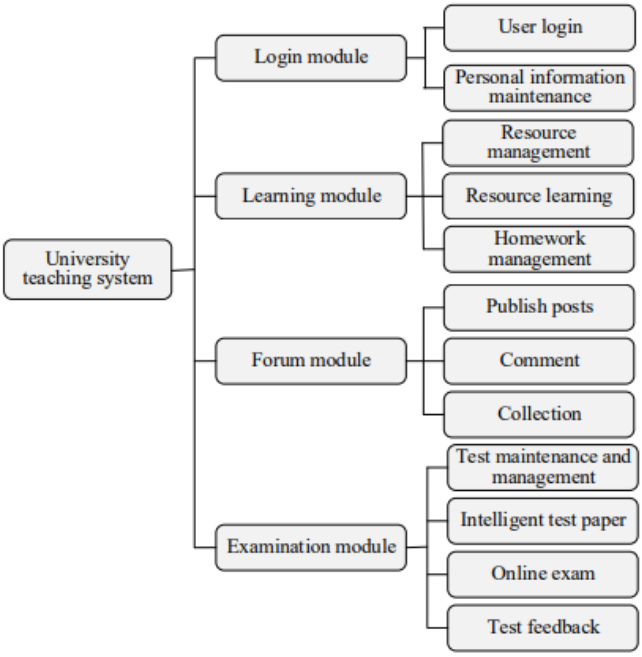


图 3.高校教学系统功能结构图

学习模块：学习功能包括三个子模块：资源管理、资源学习和作业管理。教师可以使用资源管理子模块维护课程信息，上传教学课件、视频等学习资源；学生可以使用资源学习子模块浏览教学资源，下载学习；作业管理子模块提供作业发布、作业提交和作业审核三个子功能

论坛模块：论坛模块主要是为师生提供一个互动平台。教师和学生可以通过论坛模块发布消息，评论和回复帖子，并将有趣或有用的帖子添加到书签中以供将来查看。

测试模块：测试模块由四个子模块组成：测试维护和管理、自动生成试卷、在线测试和测试反馈。在该模块中，教师可以添加和修改测试数据库；学生可以根据学习进度或自己的情况选择测试范围。系统根据学生的要求自动生成试卷。然后，学生完成自测，并检查考试成绩和试卷分析

3.3 主要系统功能模块的详细设计与实现

系统登录模块的设计与实现：高校教学系统由学生、教师和管理员三类用户组成。管理员拥有系统的所有功能。他们可以根据教师编号和学生编号设置学生和教师的角色和权限，以及用户权限和初始密码。当学生和老师的用户第一次使用系统时，他们根据给定的用户名和密码登录，并在成功登录后查看不同的界面。但他们可以根据自己的情况更改个人信息，包括修改密码、上传个人资料照片等。其他人不能进入系统，除非提交注册信息并获得批准。图 4 显示了系统登录过程。图 5 显示了登录界面。

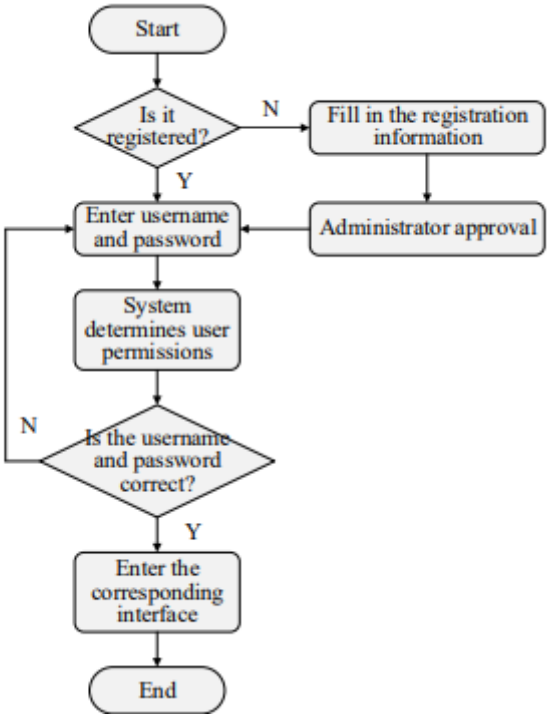


图 4.系统登录过程

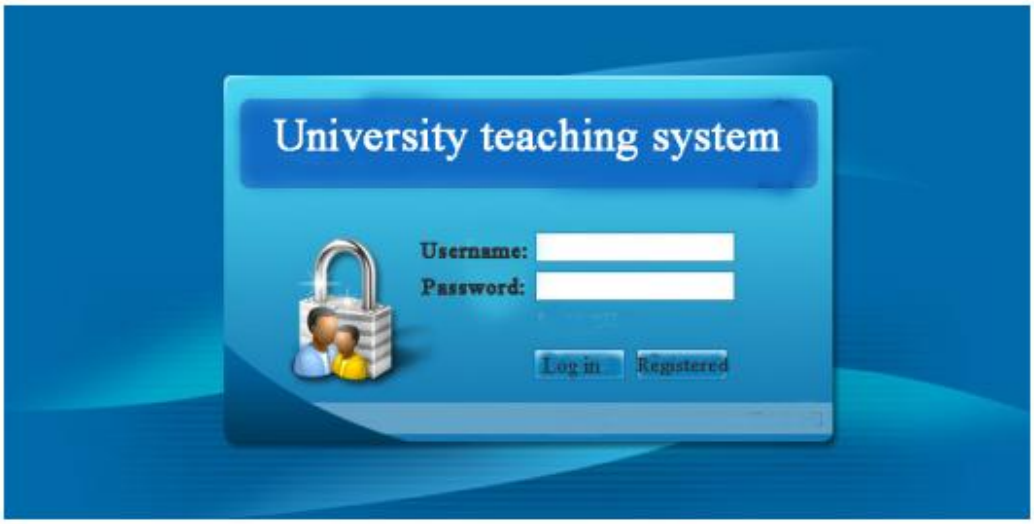


图 5. 系统登录界面

测试模块的设计与实现：输入正确的用户名和密码后，学生可以访问学生的主页。通过此页面，他们可以查看课程的通知公告和作业状态，并使用导航栏访问相应的模块链接。

点击在线考试按钮，学生进入在线考试功能模块，根据自己的学习情况选择试题类型和试题范围。

然后，系统将根据学生的要求自动组织试卷，并向他们呈现试题，而学生将在规定时间内完成并提交测试。试卷由系统自动评分，相关考试结果和试卷信息也反馈给学生自学。图 6 为在线测试流程图，图 7 为在线测试界面。

教学资源管理子模块的设计与实现：教学资源管理是学习模块的一个子模块。成功登录后，教师可以通过此模块添加、修改或删除课程信息和相关资源。当点击教学资源上传界面（图 8）时，教师可以通过拖放将编辑后的文本或视频添加到上传界面。根据教学资源的主题和具体类型，可以选择并上传相关资源供学生学习。对于试题库，教师可以批量上传，也可以根据实际需要逐一添加。图 9 显示了教师添加试题的界面。

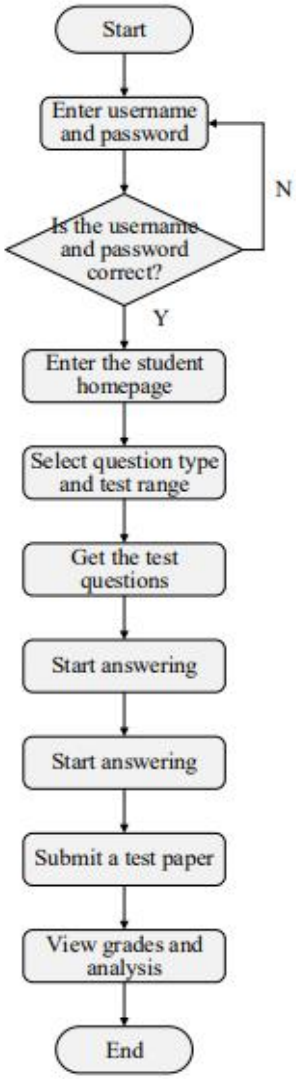


图 6.在线考试流程图

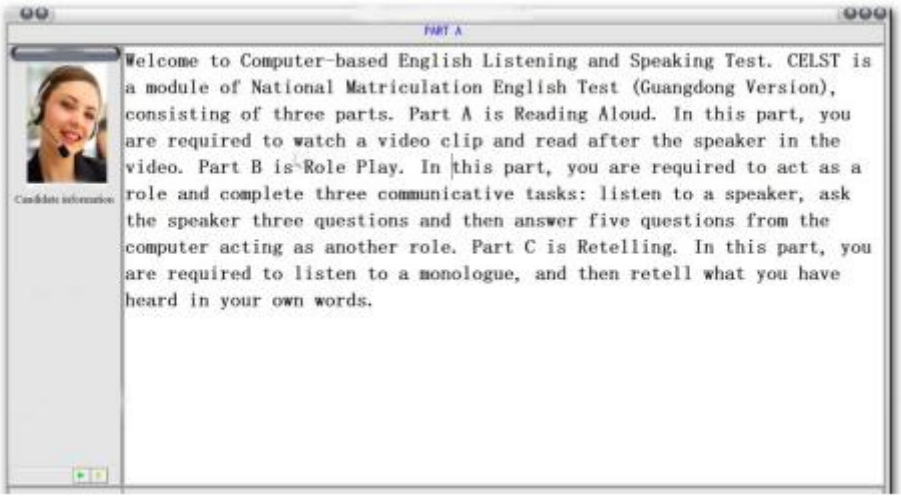


图 7.在线测试界面



图 8.上传教学资源界面



图 9.试题添加界面

3.4 系统测试

为了保证系统运行质量，及时发现系统缺陷，采用黑盒测试方法对系统稳定性和功能可扩展性进行测试。测试结果表明，系统登录界面、课程界面、视频播放界面、题库界面、在线测试界面、测试反馈界面等能够正常运行，满足用户需求

4 结论

本文在国内外教学系统研究的基础上，研究了基于 Vue.js 框架的高校教学系统的设计与实现。具体结论如下：

•从高校教学的实际需求出发，从学生、教师和管理者三个方面分析了系统的功能要求。基于此，系统分为四个模块：登录功能、学习功能、论坛功能和测试功能

•以系统登录模块、在线考试模块和教学资源管理子模块为例，阐述了系统主要功能模块的设计与实现。

黑盒测试方法用于测试系统的稳定性和功能可扩展性。测试结果表明，该系统能够正常运行，满足用户需求。

5 致谢

河南省教师教育课程改革项目：大数据背景下中小学教师网络培训模式（编号：2018-JSJYYB-029）

河南省教师教育课程改革项目：“互联网+教育”模式下中小学教师信息化课

程改革研究（NO.2019-JSJYYB-118）。

6.参考文献

[1] Wang, H. C., Hsu, C. W. (2006). Teaching-material design center: an ontology-based sys tem for customizing reusable e-materials. Computers and Education, 46(4), 458-470. https://doi.org/10.1016/j.compedu.2005.09.005
[2] Chen, H. Y., Liu, K. Y. (2008). Web-based synchronized multimedia lecture system design for teaching/learning chinese as second language. Computers and Education, 50(3), 693- 702. https://doi.org/10.1016/j.compedu.2006.07.010
[3] Craig, K., Stolfi, F. (2002). Teaching control system design through mechatronics: aca demic and industrial perspectives. Mechatronics, 12(2), 371-381. https://doi.org/10.10 16/s0957-4158(01)00076-9
[4] Ye, Z., Hua, C. (2012). An innovative method of teaching electronic system design with psoc. IEEE Transactions on Education, 55(3), 418-424. https://doi.org/10.1 109/te.2011.2181994
[5] Wang, Y. H., Lo, H. C. (2013). Teaching quality enhancement using value-chain instruc tion system design. Quality and Quantity, 47(6), 3303-3318. https://doi.org/10.10 07/s11135-012-9720-5
[6] Pu, Y. H., Chiu, P. S., Chen, T. S., Huang, Y. M. (2015). The design and implementation of a mobile library app system. Library Hi Tech, 33(1), 15-31. https://doi.org/10.1108/lht 10-2014-0100
[7] Glover, R. H., Krotseng, M. V. (1992). Design and implementation of a decision support system for institutional advancement. Journal of Computing in Higher Education, 3(2), 99- 120. https://doi.org/10.1007/bf02942358
[8] Hora, M. T., Holden, J. (2013). Exploring the role of instructional technology in course planning and classroom teaching: implications for pedagogical reform. Journal of Compu ting in Higher Education, 25(2), 68-92. https://doi.org/10.1007/s12528-013-9068-4
[9] Chang, B., Cheng, N. H., Deng, Y. C., Chan, T. W. (2007). Environmental design for a structured network learning society. Computers and Education, 48(2), 234-249. https://doi.org/10.1016/j.compedu.2004.11.008
[10] Wei, Q. L., Chang, T. S. (2011). Optimal system design series-network dfa models. Jour nal of the Operational Research Society, 62(6), 1109-1119. https://doi.org/10. 1057/jors.2010.45
[11] Zhang, H., Xu, J. L., Cheng, X. (2016). Interactive educational software for teaching water distribution networks design. Computer Applications in Engineering Education, 24(4), 615-628. https://doi.org/10.1002/cae.21736
[12] Weber, Erwin, W. (1969). Modification and use of nasap to aid in teaching network de sign. IEEE Transactions on Education, 12(4), 232-234. https://doi.org/10.1 109/te.1969.4320511
[13] Baylari, A., Montazer, G. A. (2009). Design a personalized e-learning system based on item response theory and artificial neural network approach. Expert Systems with Applica tions, 36(4), 8013-8021. https://doi.org/10.1016/j.eswa.2008.10.080
[14] Linge, N., Parsons, D. (2006). Problem-based learning as an effective tool for teaching computer network design. IEEE Transactions on Education, 49(1), 5-10.

https://doi.org/10.1109/te.2005.852600
[15] Li, Y. X., Lin, L., Qiang, W. (2013). Research and development of the integrated man agement system of the college sports based on the network. IEEE, 440-443. http:// doi.org/10.1109/isdea.2013.505
[16] Tripathy, S. C., Durgaprasad, G. S. S. S. K. (1980). The micro-machine as a teaching aid to power system simulation study. International Journal of Electrical Engineering Educa tion, 17(1), 39-50. https://doi.org/10.1177/002072098001700110
[17] Mcwilliams, A. T. (2010). Introducing expert systems to medical students using esta, ex pert system teaching aid. Medical Education, 22(2), 99-103. https://doi.org/10.11 11/j.1365-2923.1988.tb00418.x
[18] Hsu, Y. Y., Yang, C. C., Su, C. C. (2002). A personal computer based interactive software for power system operation education. IEEE Transactions on Power Systems, 7(4), 1591- 1597. https://doi.org/10.1109/59.207385

7 作者

宋俊辉毕业于中国地质大学（武汉），获硕士学位，现任信阳师范大学教师。主要研究方向是信息管理和网络建模优化。

华协毕业于同济大学，获硕士学位，现任信阳职业技术学院教师。主要研究方向是网络安全、网络建模和优化。

张敏是信阳师范大学计算机与信息技术学院的一名大学生

Design and Implementation of a Vue.js-Based College Teaching System

https://doi.org/10.3991/ijet.v14i13.10709

Junhui Song (✉), Min Zhang
Xinyang Normal University, Xinyang, China
85141727@qq.com

Hua Xie
Xinyang Vocational and Technical College, Xinyang, China

Abstract—The proliferation of Internet information technology has fundamentally changed the way of learning. It is now a research hotspot to improve teaching effect in colleges with Internet information technology. Starting from the actual needs of college teaching, this paper designs a college teaching system based on various Internet information technologies. Specifically, the front end was developed under the lightweight progressive Vue.js framework, which relies on the model–view–view model (MVVM); the overall structure of the system was set up based on the browser-server (B/S) architecture; the system functions were realized through HTML5, Node.js and database technology; the compatibility between mobile terminal and desktop was realized under Bootstrap. The system test shows that the Vue.js-based college teaching system operated stably, achieved the design goals and satisfied user demand. The research findings shed important new light on modernization and quality of college teaching.

Keywords—Vue.js framework, HTML5, teaching system, design and implementation

1 How to Work with This Template

1.1 Applying the styles to an existing paper

At present, the Internet and information technology have become indispensable technical tools in people's lives. Integrating the Internet and information technology into education has greatly promoted the teaching information reform in colleges and universities. In the face of the embarrassing situation that the traditional classroom teaching cannot meet the needs of college students under the information environment, colleges and universities have established a college teaching system based on LAN or Internet to assist traditional classroom teaching, which could provide a platform for teacher-student learning exchange and thus improve teaching effect [1].

Foreign information technology and Internet technology have developed earlier than in China. Therefore, foreign research and development on teaching systems is relatively mature. The United States has the most extensive network education, e.g., in Massachusetts Institute of Technology (MIT), the number of shared on-line courses had reached 2000 by the end of 2006 [2]. Blackboard [3] is a teaching platform developed by American companies; it is also the most widely used in the world, providing teachers with a strong curriculum development and teaching environment, and sudents with a good virtual learning environment. Sakai [4] is an open source platform of collaborative learning jointly developed by multiple universities in the United States, with the purpose of integrating the existing components and tool modules of colleges and universities to build an online teaching management. Moodle [5] is a teaching aid system developed by Australian teachers based on constructivist theory consisting of multiple modules such as resources, forums, curriculum management etc, and has been recognized by many institutions. Despite the later start of the domestic college teaching system, it has been developing rapidly. The 4A network teaching platform [6] was developed the earliest in China. This platform is composed of teaching resource management, curriculum coordination development tools, the teaching management system, and the teaching support system, which increases the teacher-students interaction and greatly mobilizes students' interest in learning. Qinghua Online established by Tsinghua University is the most mature one among the network platforms established by various universities [7]. The VClass [8] developed by Beijing Normal University has solved many prolmens in tradition teaching, and it can also provide services for distance learning, making it possible to achieve educational resource balance. With the popularity of Internet information technology and smart phones, college students have become accustomed to acquiring knowledge and learning related skills through the Internet. The collge teaching system in China has developed rapidly, but there still exist the problems such as different development levels, untimely updates, incomplete curriculum structure, and single function etc. [9].

By analysing the development status and research situation of colleges teaching systems at home and abroad, this paper attempts to study the design and implementation of college teaching system using the Vue.js framework. To this end, it expounds the Vue.js framework, HTML5, Node.js and other related technologies involved in the system development process. Then, starting from the actual teaching needs, it analyses the various requirements of the system, determines its overall architecture, discusses about the key modules in detail, and demonstrates the specific system implementation. Finally, the system test results show that the Vue.js-based college teaching system operated stably, achieved the design goals and satisfied user demand.

2 System Related Technology

2.1 Vue.js framework

Vue.js [10] is the best lightweight front-end framework based on MVVM mode in Web applications. Figure 1 shows the schematic diagram of MVVM mode architec-

ture, in which ViewModle, as a middleware, is responsible for communication between functions and data [11].

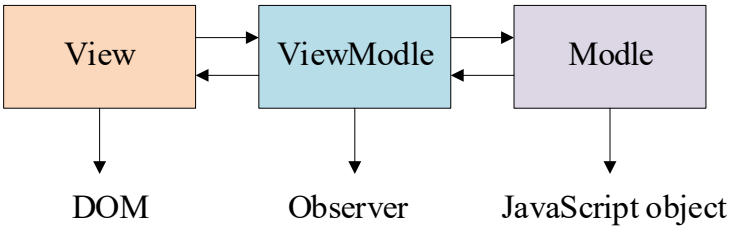


Fig. 1. MVVM mode architecture diagram

The Vue.js core library focuses only on the functional layer and is therefore not a versatile framework. Vue.js can implement responsive two-way data binding through a simple API and quickly build user interfaces [12-13]. In Vue.js, any type of application interface can be abstracted into a component tree. Figure 2 shows a schematic diagram of the component system. If the component is abstracted reasonably, several small components can be reused to build a large system, so that the repetitive development shall be reduced [14].

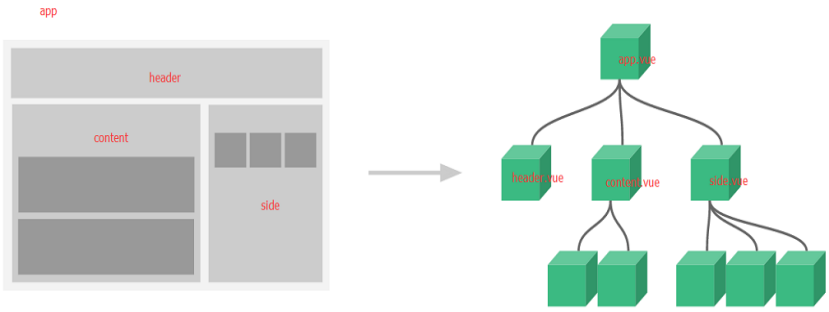


Fig. 2. Vue.js component system schematic

2.2 HTML5

Hypertext markup language (HTML) consists of the "head" and "body". It is the standard mark-up language to mark each part of a web page by the mark symbols [15]. HTML is widely used because it allows browsers to easily access new web pages by clicking on a URL pointer (hyperlink).

HTML5 [16] is the latest generation of HTML standards. Based on HTML, it has more new useful features such as performance and integration, web multimedia, local storage, semantic features and so on. It has been the biggest leap of web development standards in the recent ten years.

2.3 Node.js

Node.js is a development platform that allows JavaScript to run on the server side based on the high-performance, open source runtime environment of JavaScript [17]. Its essence is an event-based framework that prevents the secondary triggering of the Web requests through event-driven mechanisms. This paper uses Node.js event-driven, asynchronous blocking I/O, single-threaded features for system back-end design [18].

3 Design and Implementation of College Teaching System Based on Vue.js Framework

3.1 System requirements analysis

The functional requirements analysis and non-functional requirements analysis of the system are important parts of software development. Functional requirements analysis is to determine the main functions and module division of the system. The non-functional requirements analysis is based on the users' demand and experience to ensure the safety, stability, easy maintenance and scalability of the system. This paper focuses on the functional requirements analysis of the system.

The college teaching system is designed to make up for the limitations of traditional classroom teaching with time and space constraints and the lack of interactive communication between teachers and students so that the students can conduct pre-study and after-class review according to their own time schedule and practical ability. It also builds an interactive platform for teachers and students communication. Students and teachers are the main service objects of the system. Besides, the administrator is responsible for ensuring the normal operation of the system through comprehensive management. Therefore, this paper analyses the functional requirements of the system from the three aspects of students' learning needs, teachers' teaching needs and administrator's management needs.

Students: After logging in, students can modify their own information, browse the course list, select the course, watch the teaching resources provided by the teacher, courseware and other teaching resources, learn online or by downloading, view the teacher's published course announcement, assignment announcement and related messages. They can also conduct online tests for selected courses, view grades and paper analysis, and communicate instantly with teachers and other students through the forum.

Teacher: After logging in, the teachers can modify his own user information, create classes and courses, manage class information, publish relevant teaching resources and announcements, arrange student assignments, maintain test questions, view students' homework and online test analysis to understand the students' learning situation, communicate with students through the forum and answer their questions.

Administrator: After logging in, the administrator can assign the user role, modify the information of the students and teachers, and manage the information of each function module in the system.

3.2 System function module division

According to the analysis of system requirements, the system is divided into four modules: login function, learning function, forum function and test function, as shown in Figure 3.

Login module: Students, teachers and system administrators can register and log in through the system login function. After logging in, they can modify the personal information, click the corresponding function button on the system homepage to enter each function module, and view the comment content and their own bookmarks.

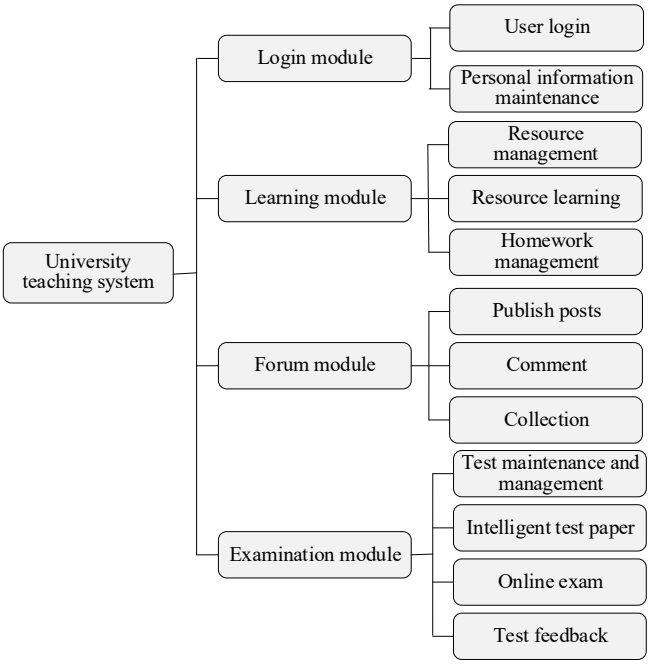


Fig. 3. Functional structure diagram of college teaching system

Learning module: The learning function includes three sub-modules: resource management, resource learning and assignment management. Teachers can use the resource management sub-module to maintain course information, upload teaching courseware, video and other learning resources; students can use the resource learning sub-module to browse teaching resources, download them for learning; the assignment management sub-module provides three sub-functions of assignment publishing, assignment submission and assignment review.

Forum module: The forum module is mainly to provide an interactive platform for teachers and students. Teachers and students can post messages through the forum module, comment and reply to posts, and add the interesting or useful posts into the bookmarks for future viewing.

Test module: The Test module consists of four submodules: test maintenance and management, automatic test paper generation, online test, and test feedback. In this module, the teacher can add and modify the test database; the students can select the test range according to the learning progress or their own situation. The system performs the automatic test paper generation according to the student's requirements. Then, the student completes the self-test, and checks the test score and test paper analysis.

3.3 Detailed design and implementation of main system function module

Design and implementation of system login module: The college teaching system consists of three types of users: students, teachers and administrators. The administrators have all the functions of the system. They can set the roles and permissions of students and teachers, as well as the user rights and initial passwords according to the teacher number and student number. When the users of student and the teacher use the system for the first time, they log in according to the given username and password, and view the different interface after the successful logging in. But they can change their personal information, including password modification, profile photo upload etc. according to their own situation. Others cannot enter the system, unless the registration information is submitted and then approved. Figure 4 shows the system login process. Figure 5 shows the login interface.

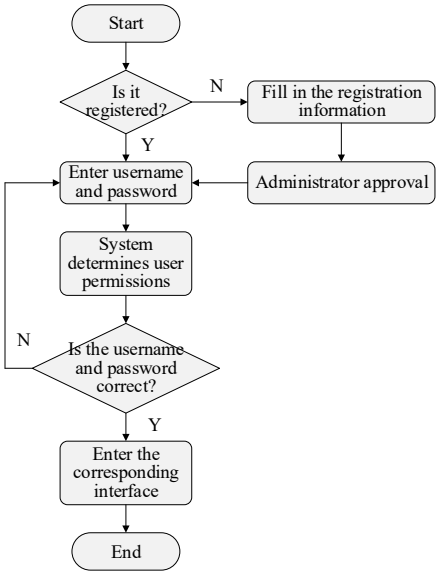


Fig. 4. System login process



Fig. 5. System login interface

Design and implementation of test module: After entering the correct username and password, the students get access to the student's main page. From this page, they can view the notice announcement and assignment status of the course, and use the navigation bar to access the corresponding module link.

Clicking the online test button, the students enter the online test function module, and choose the type of test questions and the scope of test questions according to their own learning conditions.

Then, the system will automatically organize the test paper according to the requirements of the students, and present the test questions to them, while the students will complete and submit the test within the specified time. The test paper is automatically scored by the system, and the related test results and test paper information are also fed back to the students for self-study. Figure 6 shows the online test flow chart, and Figure 7 shows the online test interface.

Design and implementation of teaching resource management sub-module: Teaching resource management is a sub-module of the learning module. After successful logging in, the teachers can add, modify or delete the course information and related resources through this module. When clicking for uploading interface of the teaching resource (Figure 8), the teacher can add the edited text or video to the uploading interface by drag-and-drop. Based on the subject and specific type of the teaching resource, the related resource can be selected and uploaded for the students' learning. For the test question bank, the teacher can make uploading in batches, or add them one by one according to actual needs. Figure 9 shows the interface for the teacher to add test questions.

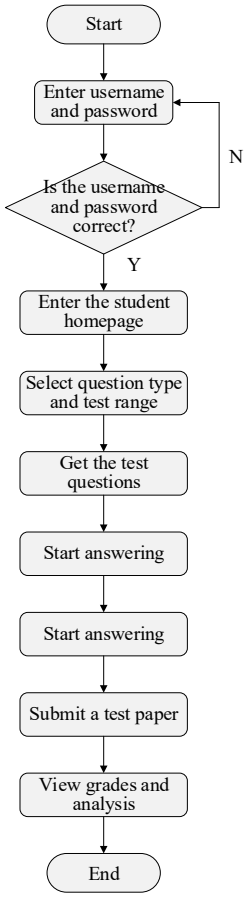


Fig. 6. Online exam flow chart

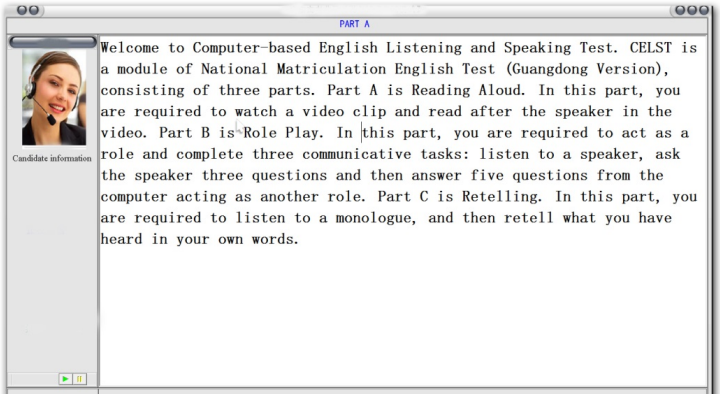


Fig. 7. Online test interface



Fig. 8. Upload teaching resource interface

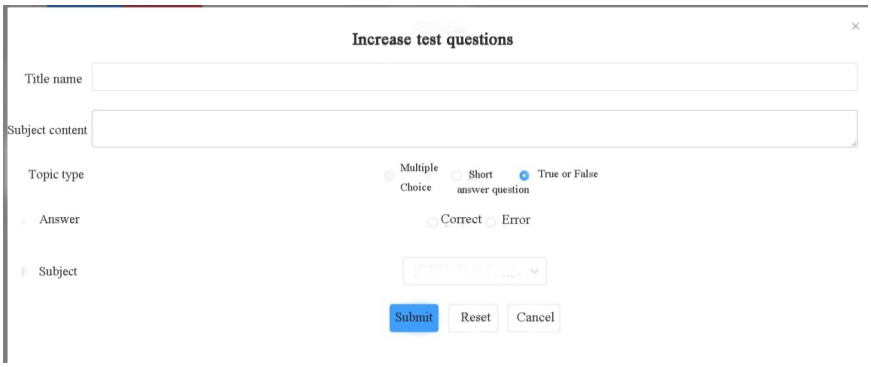


Fig. 9. Test question add interface

3.4 System test

In order to ensure the system operation quality and find the system deficiency in a timely manner, the black-box testing method was used to test the system stability and the function scalability. The test results show that the system login interface, course interface, video play interface, question bank interface, online test interface, and test feedback interface etc. can run normally and meet user demands.

4 Conclusion

Based on the teaching system researches at home and abroad, this paper studies the design and implementation of college teaching system based on Vue.js framework. The specific conclusions are as follows:

- Starting from the actual needs of college teaching, the functional requirements of the system from the three aspects of students, teachers and administrators were analysed. Based on this, the system is divided into four modules: login function, learning function, forum function and test function.

- Taking the system login module, online test module and teaching resource management sub-module as examples, the design and implementation of the main system functional modules were elaborated.

The black-box testing method was used to test the system stability and the function scalability. The test results show that the system can run normally and meet the needs of users.

5 Acknowledgement

Henan Province Teacher Education Curriculum Reform Project: Network Training Model for Primary and Secondary School Teachers under the Background of Large Data (NO. 2018-JSJYYB-029)

Henan Province Teacher Education Curriculum Reform Project: Research on the curriculum reform of teachers' Informatization in primary and secondary schools under the mode of "Internet + education" (NO.2019-JSJYYB-118).

6 References

[1] Wang, H. C., Hsu, C. W. (2006). Teaching-material design center: an ontology-based system for customizing reusable e-materials. Computers and Education, 46(4), 458-470. https://doi.org/10.1016/j.compedu.2005.09.005

[2] Chen, H. Y., Liu, K. Y. (2008). Web-based synchronized multimedia lecture system design for teaching/learning chinese as second language. Computers and Education, 50(3), 693-702. https://doi.org/10.1016/j.compedu.2006.07.010

[3] Craig, K., Stolfi, F. (2002). Teaching control system design through mechatronics: academic and industrial perspectives. Mechatronics, 12(2), 371-381. https://doi.org/10.1016/s0957-4158(01)00076-9

[4] Ye, Z., Hua, C. (2012). An innovative method of teaching electronic system design with psoc. IEEE Transactions on Education, 55(3), 418-424. https://doi.org/10.1109/te.2011.2181994

[5] Wang, Y. H., Lo, H. C. (2013). Teaching quality enhancement using value-chain instruction system design. Quality and Quantity, 47(6), 3303-3318. https://doi.org/10.1007/s11135-012-9720-5

[6] Pu, Y. H., Chiu, P. S., Chen, T. S., Huang, Y. M. (2015). The design and implementation of a mobile library app system. Library Hi Tech, 33(1), 15-31. https://doi.org/10.1108/lht-10-2014-0100

[7] Glover, R. H., Krotseng, M. V. (1992). Design and implementation of a decision support system for institutional advancement. Journal of Computing in Higher Education, 3(2), 99-120. https://doi.org/10.1007/bf02942358

[8] Hora, M. T., Holden, J. (2013). Exploring the role of instructional technology in course planning and classroom teaching: implications for pedagogical reform. Journal of Computing in Higher Education, 25(2), 68-92. https://doi.org/10.1007/s12528-013-9068-4

[9] Chang, B., Cheng, N. H., Deng, Y. C., Chan, T. W. (2007). Environmental design for a structured network learning society. Computers and Education, 48(2), 234-249. https://doi.org/10.1016/j.compedu.2004.11.008

[10] Wei, Q. L., Chang, T. S. (2011). Optimal system design series-network dfa models. Journal of the Operational Research Society, 62(6), 1109-1119. https://doi.org/10.1057/jors.2010.45
[11] Zhang, H., Xu, J. L., Cheng, X. (2016). Interactive educational software for teaching water distribution networks design. Computer Applications in Engineering Education, 24(4), 615-628. https://doi.org/10.1002/cae.21736
[12] Weber, Erwin, W. (1969). Modification and use of nasap to aid in teaching network design. IEEE Transactions on Education, 12(4), 232-234. https://doi.org/10.1109/te.1969.4320511
[13] Baylari, A., Montazer, G. A. (2009). Design a personalized e-learning system based on item response theory and artificial neural network approach. Expert Systems with Applications, 36(4), 8013-8021. https://doi.org/10.1016/j.eswa.2008.10.080
[14] Linge, N., Parsons, D. (2006). Problem-based learning as an effective tool for teaching computer network design. IEEE Transactions on Education, 49(1), 5-10. https://doi.org/10.1109/te.2005.852600
[15] Li, Y. X., Lin, L., Qiang, W. (2013). Research and development of the integrated management system of the college sports based on the network. IEEE, 440-443. http://doi.org/10.1109/isdea.2013.505
[16] Tripathy, S. C., Durgaprasad, G. S. S. S. K. (1980). The micro-machine as a teaching aid to power system simulation study. International Journal of Electrical Engineering Education, 17(1), 39-50. https://doi.org/10.1177/002072098001700110
[17] Mcwilliams, A. T. (2010). Introducing expert systems to medical students using esta, expert system teaching aid. Medical Education, 22(2), 99-103. https://doi.org/10.1111/j.1365-2923.1988.tb00418.x
[18] Hsu, Y. Y., Yang, C. C., Su, C. C. (2002). A personal computer based interactive software for power system operation education. IEEE Transactions on Power Systems, 7(4), 1591-1597. https://doi.org/10.1109/59.207385

7 Authors

Junhui Song graduated from China University of Geosciences (Wuhan) with a master's degree, is currently a teacher of Xinyang Normal University. Her main research direction is information management and network modeling optimization.

Hua Xie graduated from Tongji University with a master's degree, is currently a teacher of Xinyang Vocational and Technical College. His main research direction is network security, network modeling and optimization.

Min Zhang is a college student in the School of Computer and Information Technology of Xinyang Normal University

Article submitted 2019-03-19. Resubmitted 2019-04-24. Final acceptance 2019-04-27. Final version published as submitted by the authors.